Problem Set #5 - Answer Sheet

1. $X + 2Y \sim N(3,10)$

$X \sim N(\mu = 1, \sigma^2 = 2)$

$2Y \sim N(\mu = 2, \sigma^2 = 8)$

According to the sum of independent normals

$X + 2Y \sim N(3,10)$

2. See the code file

3. $P(\sum_{i=1}^{80} X_i \leq 300)$

Let X be a ramdon variable represents the dice point of each roll. Then $X \sim Uni(1,2,3,4,5,6)$, where the distribution is discrete, $\mu = 3.5$ and $\sigma^2 = 2.917$. According to the central limit therom, $\sum_{i=1}^{n} X_i \sim N(n\mu, n\sigma^2)$.

Therefore $p = P(\sum_{i=1}^{80} X_i \leq 300)$

4.

a. $P(\sum_{i=1}^{20} X_i \leq 950)$

$\mu_1 = 50, \sigma^2 = 100$. According to the central limit therom, $\sum_{i=1}^{n} X_i \sim N(n\mu_1, n\sigma_1^2)$

Therefore, $p_1 = P(\sum_{i=1}^{20} X_i \leq 950)$

b. $P(\sum_{i=1}^{20} X_i \leq 950)$

$\mu_2 = 52, \sigma^2 = 200$. According to the central limit therom, $\sum_{i=1}^{n} X_i \sim N(n\mu_2, n\sigma_2^2)$

Therefore, $p_2 = P(\sum_{i=1}^{20} X_i \leq 950)$

c. $P(X - Y < 0)$

Transform the question to asking the probability that the probability that the run time of Program A minus the run time of Program B is less than 0. Let X, Y represent that run time of Program A and Program B, repectively.

Because $X \sim N(1000,2000)$, $Y \sim N(1050,4000)$, and X and Y are independent. According to the sum of independent normals, $X - Y \sim N(-50,2000)$.

Therefore $p_3 = P(X - Y < 0)$

5.

a. 5.95

Let X be a random variable represents the money that each people donates. According to the law of total expectation, $E[X] = \sum_{y} E[X \mid Y = y]P(Y = y) = 5.95$

b. 23.5475

$Var(X) = E[X^2] - E^2[X]$,

$E^2[X] = 35.4025$

$E[X^2] = \sum_{y} E[X^2 \mid Y = y]P(Y = y) = 58.95$

Therefore, $Var(X) = 23.5475$

c. $S \sim N(5.95,23.5475)$

Let S be a random variable represents the total amount of money earned. Because $S = \sum_{i=1}^{50} X_i$,

then according to the central limit theorem, $S \sim N(297.5,1177.375)$

d. 0.482

$$p = 1 - F(s = 350) = 0.482$$

6.

a. 6

Let

X: A random variable represents the dice point of each roll

T: A bernoulli random represents whether the dice point of each roll is greater or equal to 3

Accoring to the law of total expectations

$$E[X] = \sum_{y} E[X \mid T = t]P(T = t)$$

where

$$E[X \mid T = 0] = 1.5$$

$$E[X \mid T = 1] = 4.5$$

Therefore

$$E[X] = 6$$

b. 2.5

$$E[Y] = 1 + \frac{1}{4} \times 2 + \frac{1}{2} \times \frac{1}{4} \times 2 + (\frac{1}{2})^2 \times \frac{1}{4} \times 4 + \ldots + (\frac{1}{2})^{n-1} \times \frac{1}{4} \times (n+1) = 1 + \sum_{n=1}^{\infty} (\frac{1}{2})^{n-1} \frac{1}{4}(n+1) = 2.5$$

7.

Let

X: The number of people infected by this person

Y: The number of people made contact to this person

According to the law of total expectation

$$E[X] = E[X \mid Y = 1]P(Y = 1) + E[X \mid Y = 2]P(Y = 2) + \ldots + E[X \mid Y = n]P(Y = n) = \sum_{n=1}^{100} E[X \mid Y = n]P(Y = n)$$

where $E[X \mid Y = n] = np = 0.01n$

$$P(Y = n) = \binom{100}{n}(0.25)^n(0.75)^{100-n}$$ b

8.